

Pico 2 W -> AM26C31 RS-422 driver -> CN0035 RJ45 (Falcon differential receiver, dumb mode)

Power: USB into the Pico only. VBUS (pin 40) feeds the 5 V net; AM26C31 G and -G both tied low (-G low = always enabled).

RJ45 pairs: port1 = pins 1(+),2(-) port2 = 3(+),6(-) port3 = 4(+),5(-) port4 = 7(+),8(-). Y = +, Z = -.

Driver map: GP5->2A->port1, GP3->4A->port2, GP4->3A->port3, GP2->1A->port4 (main.py DATA_PINS = (5, 3, 4, 2)).

Rev B for JLCPCB assembly: U1 Pico 2W C42394205 | U2 AM26C31IDR C34923 | J1 RJ45 C385834 | C1 C49678 | C2 C15850 | J2 C32713268 | H1-H4 M3

